Optima Academy Online
AN EDUCATION EXPERIENCE COMPANY

Full Year · Grades 9–12

Algebra 2 Honors

COURSE SYLLABUS · 2026–2027

FACILITATOR — EDIT THESE FIELDS

Replace the text inside each [[bracket]] with your own information, then delete the brackets. Delete this reminder line when you are finished.

FACILITATOR
[[Your name]]

EMAIL
[[your.email@optimaed.com]]

OFFICE HOURS
[[e.g., Tue & Thu 3:00–4:00 PM ET, plus by appointment]]

LIVE CLASS MEETINGS (IF APPLICABLE)
[[e.g., Mon & Wed 10:00–11:00 AM ET on the VR campus]]

BEST WAY TO REACH ME
[[e.g., Canvas Inbox; I reply within 24 hours on school days]]

FRONT OFFICE
239.673.2801

FLORIDA COURSE
Algebra 2 Honors · #1200340

GRADE LEVEL
Grades 9–12 (High School)

CREDIT & LENGTH
1.0 Mathematics credit · Full-year course

FORMAT
On-Demand (also offered live on the VR campus)

About this syllabus. This is the plan for the course as designed. It is a full-year Honors course of 14 modules plus two Honors modules; On-Demand scholars pace through the modules across the four quarters of their enrolled year. On-Demand and live-school sections share the same content and expectations — exact pacing, scheduled dates, and a few activities may vary from section to section and term to term. Your facilitator will confirm the specifics for your section.

COURSE OVERVIEW

Algebra 2 Honors extends the algebra you have already met into the full landscape of functions and the complex number system. Instructional time emphasizes six areas: (1) developing an understanding of the complex number system, including complex numbers as roots of polynomial equations; (2) extending arithmetic with algebraic expressions to include polynomial division and radical and rational expressions; (3) graphing and analyzing polynomial, absolute value, radical, rational, exponential, and logarithmic functions; (4) extending systems of equations and inequalities to include non-linear relationships; (5) building functions using compositions, inverses, and transformations; and (6) developing an understanding of probability. Each concept pairs a short interactive lesson with independent practice, a mid-module check, and an immersive activity on Optima's VR campus, and every module builds toward a summative assessment. As an Honors course, it carries added rigor — deeper reasoning, richer contexts, and additional Honors-only topics such as the Binomial Theorem, sequences, piecewise functions, probability, and matrices.

WHAT SCHOLARS WILL LEARN TO DO

- Work within the complex number system and use complex numbers as roots of polynomial equations.
- Add, subtract, multiply, and divide polynomials, and divide by long and synthetic division.

- Factor and solve quadratic and higher-degree polynomial equations and inequalities.
- Graph and analyze polynomial, rational, and radical functions and their key features.
- Simplify and operate on rational and radical expressions and rational exponents.
- Work with logarithms and solve exponential and logarithmic equations.
- Graph and transform exponential and logarithmic functions.
- Compare, combine, compose, and invert functions of many types.
- Model and solve absolute value, piecewise, and systems problems, including non-linear systems and constraints.
- Reason about probability and (Honors) work with sequences and matrices.

SEMESTER 1 AT A GLANCE

Semester 1 covers Modules 1–7 and the Semester 1 Honors module. Approximate lengths are a guide for pacing; your section may vary.

MODULE	FOCUS	APPROX.
Module 1 Square Roots & Complex Numbers	Square roots, imaginary and complex numbers, and operations on complex numbers.	≈ 8 days
Module 2 Quadratic Equations in One Variable	Polynomials, factoring quadratics, and solving quadratic equations and inequalities.	≈ 12 days
Module 3 Quadratic Functions & Inequalities in Two Variables	Graphing, writing, and transforming quadratic functions; two-variable quadratic inequalities.	≈ 12 days
Module 4 Polynomial Equations in One Variable	Operations on polynomials, division, factoring, the Remainder Theorem, and solving. Honors: the Binomial Theorem.	≈ 13 days
Module 5 Polynomial Functions in Two Variables	Graphs of polynomial functions and transformations of cubic functions.	≈ 7 days
Module 6 Rational Expressions & Functions	Operations on rational expressions, rational equations, and graphing rational functions.	≈ 10 days
Module 7 Radical Equations & Functions	Radical equations, graphing radical functions, and transforming square and cube root functions.	≈ 8 days
Semester 1 Honors Probability	Sample spaces, independent and conditional probability, and permutations and combinations.	≈ 12 days

Semester 2 continues with Modules 8–14 (rational exponents and radicals, logarithms, exponential and logarithmic functions, properties of functions, inverse functions, absolute value and piecewise functions, and systems) and the Semester 2 Honors module on Matrices.

IMPORTANT DATES — 2026–2027

Based on the board-approved academic calendar. Confirm against the current school calendar and update if the school adjusts any date.

Aug 12, 2026	First day of school	**Dec 21–Jan 1**	Winter break

Sep 7	Holiday (Labor Day)	**Jan 5, 2027**	Classes resume
Oct 9	Quarter 1 ends	**Jan 18**	Holiday (MLK Day)
Oct 12	Holiday (Columbus Day)	**Feb 15**	Holiday (Presidents' Day)
Nov 11	Holiday (Veterans' Day)	**Mar 5**	Quarter 3 ends
Nov 23–27	Thanksgiving break	**Mar 26–Apr 5**	Spring break
Dec 18	Q2 ends · fall semester ends	**May 28, 2027**	Q4 ends · last day of school

ASSESSMENT & GRADING

Your grade is built from the work you do in every module. Each mini-lesson opens with a **Prior Knowledge Check** (graded for completion) and a short **Independent Practice** (4–10 problems). Each module includes one or more **Mid-Module Checks**, a **Summative Review** (graded for completion, with unlimited attempts), and a **Module Summative Assessment**. Immersive **VR lesson activities** support the lessons throughout.

By default this course is scored on total points across all assignments. If your section weights categories, the suggested weighting below is a starting point — edit the numbers to match your gradebook.

50%
Independent practice & mid-module checks

40%
Module summative assessments

10%
Completion & VR activities

HOW THIS COURSE IS DELIVERED

This course is offered in two formats that share the same content, assignments, and expectations: an **On-Demand (asynchronous)** section you pace yourself, and a **Live (synchronous)** section that meets on a schedule on Optima's VR campus. Find the one that matches your enrollment.

ON-DEMAND (ASYNCHRONOUS) LEARNING

In the On-Demand section you move through the course at your own pace within each quarter, guided by due dates that keep you on track. All assignments for a quarter open at its start and close at its end. Because the course is On-Demand, there is no grade penalty for missing an individual due date — but any assignment not completed by the end of the quarter receives a zero, and assignments submitted in the final 14 days of a quarter receive a 10% deduction, so steady pacing matters.

Attendance is measured by assignment completion. Per school policy, a scholar who completes no assignment for ten consecutive school days, without notifying the school of an absence in advance, may be dropped from the course. Completion means a sincere attempt — real effort on real work. Rushed work, wrong or corrupted files, nonsense submissions, AI-generated filler, and word-salad responses do not count.

You are never on your own. A live facilitator grades your work, gives feedback, and holds office hours and VR support.

LIVE (SYNCHRONOUS) LEARNING

In the Live section you join scheduled class meetings on Optima's VR campus, where your teacher leads instruction in real time and you learn alongside classmates. The lessons, practice, and assessments are the same as the On-Demand course; what differs is the pacing — a Live section moves through the material together on the class schedule rather than at each scholar's own pace, and class time is also used for worked examples, questions, and guided practice.

Attendance is measured by presence and participation. You are expected to attend the scheduled live sessions, on time and ready to take part. If you must miss a class, notify the school in advance; your teacher will let you know how to make up the work and any material you missed. As in every Optima course, completing your work means a sincere attempt — real effort on real work — and assignments still open and close with each quarter.

🎮 You meet regularly with a teacher. The class gathers in VR and on Microsoft Teams for live instruction and discussion.

APPROPRIATE USE OF AI

Optima scholars are preparing for a world shaped by artificial intelligence, and learning to use these tools with good judgment is part of that preparation. You are welcome to use AI tools to support your learning — to explain a concept a different way, check your understanding of a worked example, or generate extra practice problems — unless a particular assignment says otherwise. What AI may never do is the thinking or the work an assignment asks of you: your solutions, your reasoning, and your explanations on graded work must be genuinely your own. Submitting AI-generated work as if it were your own is dishonest and counts as an insincere attempt. When you do use an AI tool on an assignment, say so briefly. Above all, build the underlying skill yourself — in mathematics the goal is for you to be able to judge whether an answer is right, which you can only do if you can do the work without it.

LEARNING IN VIRTUAL REALITY

Optima is a virtual-reality school. Alongside your Canvas coursework, you will join immersive sessions on Optima's VR campus, where you meet teachers and fellow scholars, take part in live activities, and explore mathematical ideas in ways an ordinary classroom cannot. Keep your headset charged and ready for any scheduled VR sessions, and treat the VR campus with the same courtesy and seriousness you would a physical one. If you ever cannot get into VR for a session, tell your facilitator — there is an alternative way to take part so you do not fall behind.

BEING AN OPTIMA SCHOLAR

Optima's mission is an excellent, virtue-centered education, and virtue is at the center of this course as much as any other. Every scholar is expected to practice the seven virtues that shape life at Optima — **courage, courtesy, honesty, perseverance, self-government, service, and responsibility**. In mathematics that means: do your own honest work; show your reasoning; persevere through hard problems and treat mistakes as part of learning; treat everyone you meet in, through, or because of this course with respect; come to office hours to get help and get to know your facilitator; and take responsibility for your own pacing and progress. You belong to a house and a community of scholars, not only to a course.

WHAT YOU NEED & STAYING CONNECTED

- A computer with reliable internet and access to **Canvas**.
- A **VR headset**, charged and ready for the immersive activities on Optima's VR campus.
- **Microsoft 365** (Word, Excel) with OneDrive for any documents you create and submit.
- The **Desmos** online graphing calculator (linked in the course) and a scientific or graphing calculator as your section requires.
- A notebook and pencils for working problems by hand — showing your work is part of learning the mathematics.

Staying connected. Updates about this course are posted on the course's Canvas page; school-wide updates are posted in your Homeroom Canvas course. Check both regularly. Your facilitator holds office hours and is glad to help with the material, to meet you, or to talk through your plans — come by, or reach out any time by Canvas Inbox.

Algebra 2 Honors · Florida course #1200340 · Full-year Honors course · Optima Academy Online, 2026–2027. Built from the course content (14 modules plus two Honors modules) and the board-approved 2026–2027 academic calendar. Pacing reflects a sample full-year plan; On-Demand scholars pace to their own enrolled term. School calendar dates may be adjusted by the school.